

MANUFACTURING PROCESSES LAB**II B TECH I Semester : MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6ME13	PCC	L	T	P	C	CIE	SEE	Total
		0	0	2	1	40	60	100

COURSE OVERVIEW:

This course is designed to impart hands-on practical exposure on manufacturing processes and equipment.

COURSE OUTCOMES:

The student will be able to:

1. Design and manufacture different patterns
2. Prepare various castings.
3. Operate arc welding, gas welding and resistance welding equipment
4. Perform operations such as blanking, piercing, deep drawing, extrusion, bending and other operations.
5. Use injection moulding and blow moulding equipment to produce plastic products such as water bottle, water bottle caps etc.

LIST OF EXPERIMENTS:**METAL CASTING**

1. Pattern Design and making - for one casting drawing.
2. Sand properties testing – 1 Exercise -for strength and permeability
3. Moulding Melting and Casting - 1 Exercise

II.WELDING

1. ARC Welding Lap & Butt Joint - 2 Exercises
2. Spot Welding - 1 Exercise
3. TIG Welding - 1 Exercise
4. Plasma welding and Brazing - 2 Exercises (Water Plasma Device)

III. MECHANICAL PRESS WORKING

1. Blanking & Piercing operation and study of simple, compound and progressive press tool.
2. Hydraulic Press: Deep drawing and extrusion operation.

IV. PROCESSING OF PLASTICS

1. Injection Moulding
2. Blow Moulding